Anshul Goyal

Senior Software Developer

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About

I am a seasoned software developer specializing in designing and implementing clean, efficient, and scalable systems across multiple programming languages. I am committed to delivering high-quality, reliable solutions that translate complex and ambiguous business requirements into robust, resilient, and maintainable production services.

Education

Post-Graduation 2009-2011

Centre for Digital Media, GNWC, Vancouver BC, Canada.
Masters in digital media

Graduation 2002-2006

National Institute of Technology, Rourkela, India.
Bachelors in computer science with Honors.

References

Available upon request.

Skills

Programming Languages: C, C++, C#, .NET Core, Python, Go, Java, Scala, Ruby, Rust
LLM: Claude Code, OpenAI Codex
Concepts: Multiplayer, Online Services, REST, GraphQL, gRPC, Asynchronous Programming
Big Data/Streaming: Apache Kafka, Apache Spark, Apache Flink (SQL), Apache Beam
Cloud Technology: Microsoft Azure, Amazon Web Services, Google Cloud Platform, Kubernetes
Databases: MySQL, Redis, Cassandra, PostgreSQL, Clickhouse
Tools/IDE: Microsoft Visual Studio, Perforce, Git
Platforms: PC, PS4, Xbox One, Nintendo Switch
Game Engines/Libs: Scimitar (Ubisoft's Internal Game Engine), Unity3D, UnrealEngine4 (Limited)

Professional Work Experience

Current Role

Senior Developer – Shopify, Montreal (QC), Canada, May'20 – present

Analytics (Current Project): Gained deep understanding of a large-scale streaming architecture comprising 150+ real-time streaming pipelines built with Google Cloud Dataflow in Java and Clickhouse, which powers the reports displayed on the Shopify admin analytics page. Actively contributing to the migration of these pipelines to Apache Flink using Flink SQL while identifying and addressing operational and development pain points from the existing system.

Identity Graph: Worked on building a high-throughput, low-latency backend GraphQL service using Ruby and gRPC to resolve and connect user identifiers in real-time, later worked on rewriting the same service in Rust to improve performance and reliability. Contributed to real-time data streaming pipelines using Java/Scala with Apache Flink to ingest and process high-volume event data from multiple Apache Kafka sources. Optimized Ruby-based GraphQL API by applying tracing, profiling, and concurrent execution, etc. to significantly reduce query processing time and be able to run queries efficiently across more than a single shop. These improvements eliminated reliance on cached responses, enabled delivery of more up-to-date data, and supported the successful launch of datasets powering the ads platform.

Customer Data Platform: Worked on overhauling the backend service powering the Customers section in the Shopify Admin. Designed and implemented low-latency, high-throughput services in Go with GraphQL APIs and gRPC addressing the need to segment customers. Built scalable data ingestion pipelines using Python with Apache Spark, java with Apache Flink to aggregate multi-source data from Apache Kafka for the API. Improved system reliability and reduced processing time through targeted performance optimizations, enabling the platform to handle a high number of requests more efficiently.

Previous Roles

- **Network/Online Programmer – Ubisoft, Montreal (QC), Canada, June'12 – May'20**

Proudly shipped the following titles on PC, Xbox and PlayStation: Hyperscape, Rainbow Six: Siege, Assassin's Creed Unity, Watch Dogs, Assassin's Creed IV Black Flag, Tom Clancy's Splinter Cell: Blacklist.

Hyperscape: Ported Ubisoft's C++ networking and online-services stack to iOS and Android, delivering a validated proof-of-concept that unblocked cross-platform development. I then contributed to backend integration work, implementing and hardening online service flows in C++ to prepare the game for production launch. To ensure a smooth release, I worked on improving service scalability, reliability, and resiliency across the online platform, supporting high-traffic, real-world load.

Rainbow Six Siege: Integrated critical online services including but not limited to party systems, matchmaking, session flows into the game engine using C++, ensuring platform-ready functionality for PC, Xbox, and PS4. Led the transitioning of major components of the network stack from peer-to-peer to a dedicated-server model and contributed directly to dedicated game server development.

Throughout the project, I drove overall product quality by providing extensive code reviews and strengthening the reliability and robustness of all online-related systems.

Assassin's Creed Unity: Integrated critical online services including but not limited to party systems, matchmaking, and session management into the game engine using C++, while ensuring compliance with technical requirements across PC, Xbox, and PS4. Drove backend readiness for launch by strengthening service flows, resolving platform-specific issues, and elevating overall product quality. Throughout the project, I contributed heavily to code reviews for all online-related components, improving reliability, maintainability, and consistency across the online stack.

Technology Group Online: Contributed to the design and development of shared backend services including but not limited to matchmaking, rewards, authentication using Python. I implemented scalable, fault-tolerant features, improved service performance, and participated in deployment and monitoring efforts to ensure reliability. These enhancements strengthened system resilience and delivered consistent, high-availability experiences to millions of players across all titles.

- **Server Engineer – Electronic Arts (EA), Montreal (QC), Canada, May'11 – June'12**

Proudly shipped the following titles on iPhone and iPad: FIFA Superstars iOS, Trouble Makers iOS, Restaurant City.

Developed backend services in Java to support core functionality for smartphone games. Implemented and integrated these services within the game client built using Unity3D to enable seamless communication between the client and backend systems. Delivered reliable service integrations that supported gameplay features and improved the stability of client-server interactions.

- **Sr. Subject Matter Expert/Software Engineer -- Amdocs, Limassol, Cyprus (Jul'06 – Aug'09)**

I contributed to the enhancement of existing products by devising creative solutions aligned with customer requirements and ensuring support for current production versions. My responsibilities encompassed requirement analysis, effort estimation, designing, coding and testing across diverse modules, leading to significant experience in developing large scale enterprise applications using Java, EJB, JSP and JMS.

- *Awarded a Letter of Recognition for my diligence towards work, mentoring new people and following the system methodologies efficiently & effectively at Amdocs Cyprus Pvt. Ltd in Mar'07.*
- *Awarded a Certificate of Appreciation for working responsibly under less supervision and maintaining consistency in a high level of quality and productivity at Amdocs Cyprus Pvt. Ltd in Aug'08.*

Internships

SDE (Intern) -- Big Park, Microsoft Game Studios, Vancouver (BC), Canada, May'10 – Aug'10

I contributed to the development of an unreleased game title, focusing on building prototypes featuring synchronous and asynchronous gameplay mechanics using Action Script 3.